PO11Q & PO12Q

Formulae Collection and Notation

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Table 1: Formulae for PO11Q

Statistic	Formula
Confidence Interval	$Pr(\bar{y} - t_{\alpha/2} \cdot se \leq \mu \leq \bar{y} + t_{\alpha/2} \cdot se) = 1 - \alpha$
Deviation	$d = y_i - \bar{y}$
Mean	$\bar{y} = \frac{\Sigma y_i}{n}$
	$\mu = \Sigma y P(y) = E[y]$
Position of p^th percentile	$P = (n + 1) \cdot \frac{p}{100}$
Range	$y_{range} = y_{max} - y_{min}$
Standard Deviation	$s = \sqrt{\frac{\Sigma(y_i - \bar{y})^2}{n-1}}$
Standard Error	$\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$
	$se = \frac{s}{\sqrt{n}}$
t-test	$t = \frac{\bar{y} - \mu_0}{se}$
Variance	$s^2 = \frac{\Sigma(y_i - \bar{y})^2}{n-1}$
z-score	$z = \frac{y - \mu}{\sigma}$
Cohen's *d* (Observed)	$d_{\text{obs}} = \frac{\bar{y} - \mu_0}{s}$
Cohen's *d* (True)	$d_{\text{true}} = \frac{\mu - \mu_0}{\sigma}$
Noncentrality Parameter	$\lambda = \frac{\mu - \mu_0}{\sigma_{\bar{y}}}$
	$\lambda = d_{\text{test}}\sqrt{n}$
Statistical Power	Power = $P(\text{reject } H_0 \mid H_A \text{ is true})$
	$1 - \beta$

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Table 2: Formulae for PO12Q

Statistic	Formula
Correlation Coefficients	
Covariance	$\text{cov}(x, y) = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$
Pearson's r	$r_p = \frac{\text{cov}(x, y)}{s_x s_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$
Spearman's r_s	$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$
Point-Biserial r_{pb}	$r_{pb} = \frac{\bar{x}_1 - \bar{x}_0}{s} \sqrt{\frac{n_1 n_0}{n(n-1)}}$
ϕ Coefficient	$\phi = \frac{AD - BC}{\sqrt{(A+B)(C+D)(A+C)(B+D)}}$
Correlation – Significance Testing	
Pearson (Significance)	$t_{\text{obs}} = \frac{r_p \sqrt{n-2}}{\sqrt{1 - r_p^2}}$
Spearman (Significance)	$t = \frac{r_s \sqrt{n-2}}{\sqrt{1 - r_s^2}} \quad \Bigg\} \quad t = \frac{r\sqrt{n-2}}{\sqrt{1 - r^2}}$
Point-Biserial (Significance)	$t = \frac{r_{pb} \sqrt{n-2}}{\sqrt{1 - r_{pb}^2}}$
ϕ (Significance)	$\chi^2 = n\phi^2 \quad (df = 1)$
Noncentrality Parameter (power)	$\lambda = \frac{\rho \sqrt{n-2}}{\sqrt{1 - \rho^2}}$

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Statistic	Formula	
Regression		
Adjusted R^2	$\bar{R}^2 = 1 - \frac{\sum \hat{\epsilon}_i^2 / (n-k-1)}{\sum (y_i - \bar{y})^2 / (n-1)}$	
Coefficient of Determination	$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum \hat{\epsilon}_i^2}{\sum (y_i - \bar{y})^2}$	
Estimated Variance	$\hat{\sigma}^2 = \frac{\sum \hat{\epsilon}_i^2}{n-p}$	
Estimator of β_0	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$	$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y})$
Estimator of β_1	$\hat{\beta}_1 = \frac{\sum_{i=1}^{N} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N} (x_i - \bar{x})^2}$	
Explained Sum of Squares	$\sum (\hat{y}_i - \bar{y})^2$	
Residual Sum of Squares	$\sum \hat{\epsilon}_i^2 = \sum (y_i - \hat{y}_i)^2$	
Estimated Standard Error of $\hat{\beta}_0$	$\hat{se}(\hat{\beta}_0) = \sqrt{\frac{\sum x_i^2}{n \sum (x_i - \bar{x})^2}}\ \hat{\sigma}$	$\hat{\mathrm{var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2 (\mathbf{X}'\mathbf{X})^{-1}$
Estimated Standard Error of $\hat{\beta}_1$	$\hat{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum (x_i - \bar{x})^2}}$	
Total Sum of Squares	$\sum (y_i - \bar{y})^2$	

2.1 Matrices

$$\boldsymbol{X'X} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$$

$$(\boldsymbol{X'X})^{-1} = \frac{1}{n \sum x_i^2 - \sum x_i \sum x_i} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}$$

$$\boldsymbol{X'y} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

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Table 3: Notation for PO11Q

Symbol	Explanation
d	Deviation
n	Sample Size
s	Standard Deviation
s^2	Variance
$\bar{y}$	Mean
y_i	Observation i
f	(Absolute) Frequency
cf	Cumulative (Absolute) Frequency
rf	Relative Frequency
crf	Cumulative Relative Frequency
$E[x]$	The expected value of x
μ	Mean of the Population
se	Standard Error (with s of sample)
σ	Standard Deviation of the Population
$\sigma_{\bar{y}}$	Standard Error (with σ of population)
t	t-value
z	z-value
H_0	Null Hypothesis
H_A (H_1)	Alternative Hypothesis
α	Significance Level / Probability of a Type I Error
β	Probability of a Type II Error
$1 - \beta$	Statistical Power
μ_0	Population Mean Specified by the Null Hypothesis
d_{obs}	Observed Effect Size (Cohen's d)
d_{true}	True Population Effect Size
d_{test}	Assumed Effect Size Used in Power Analysis
λ	Noncentrality Parameter
t_{crit}	Critical t-Value

Table 4: Notation for PO12Q

Symbol	Explanation
χ^2	Chi-Squared for test of independence
β	Regression Coefficient
$\hat{\beta}$	Estimated Regression Coefficient
ϵ	Error Term
$\hat{\epsilon}$	Estimated Error Term / Residual
A^{-1}	Inverse of Matrix A
A′	Transpose of Matrix A
I	Identity Matrix
R^2	R-Squared / Model Fit
σ^2	Mean Squared Error
$\hat{\sigma}^2$	Residual Standard Error
k	Number of Slope Coefficients
K	Total Number of Regression Coefficients, including the Intercept
$\bar{R}^2$	Adjusted R-Squared
log(x)	Logarithm of variable x
α_i	Regression coefficients of secondary regression models
VcV	Variance-Covariance Matrix
Ω	Is equal to $\sigma^2 I$, where σ^2 represents the mean squared error
t	Time period in time-series data